Factoring polynomials

1 Consider the expression $m^2 - 16 + 2m + 8$.

- **a** Factor $m^2 - 16$.
- **b** Factor $2m + 8$.
- **c** Using parts (a) and (b) or otherwise, factor $m^2 - 16 + 2m + 8$.

2 Factor:

- **a** $x^4 - 1$
- **b** $1 - n^4$
- **c** $18 - 8t^2$
- **d** $27t^3 - 48t$
- **e** $n^3 - 121n$
- **f** $9x^2y^2 - 16z^2w^2$
- **g** $16t^2 - 36t^4$
- **h** $5t^2 - 20t^4$
- **i** $4p^2q^2 - 81c^2d^2$
- **j** $x^2y^2z^2 - 36x^2$
- **k** $810p^2 - 10q^2$

3 Factor:

- **a** $pqr - p^2q^2r^2 - p^3q^3r^3$
- **b** $-7\left(-3x + 9\right) + 8x\left(-3x + 9\right)$
- **c** $8x\left(10y - z + w\right) - 9u\left(10y - z + w\right)$
- **d** $5y\left(y - 10\right) + 4\left(10 - y\right)$.
- **e** $8p\left(p^2 - 100\right) - 5\left(p^2 - 100\right)$

4 Factor:

- **a** $x^2 - 6x + 9 - y^2$
- **b** $x^2 + 8x + 16 - x^4$
- **c** $x^4 - x^2 - 16x - 64$
- **d** $9x^2 - 24x + 16 - 4y^2$
- **e** $u^2 - x^2 + 8xy - 16y^2$
- **f** $56xy + wx + 56yz + wz$
- **g** $4x + 45yz + 36xy + 5z$
- **h** $x^2 - x^2y + xy - x$
- **i** $x^2 - y^2 - x - y$
- **j** $m^3 - 9m - 5m^2 + 45$

5 Factor:

a $5x^2 + 30x + 40$

b $6x^2 - 36x + 48$

c $3x^2 - 12x - 36$

d $w^4 - 5w^2 - 36$

e $20x^2 - 25x - 30$

f $60x^2 - 39x + 6$

g $20x^3 + 25x^2 - 30x$

h $5x^3 + 13x^2 + 6x$

i $80x^4 + 92x^3 + 24x^2$

j $15x^2y + 50xy - 40y$

k $-12x^2y^4 - 57xy^4 + 15y^4$

l $5x^6 - 17x^3 - 12$

m $9p^2 + 24pq + 16q^2$

n $9p^2 - 30pq + 25q^2$

o $p^3 + 8p^2q + 16pq^2$

p $x^4 - 10x^3 + 24x^2$

q $-2x^4 + 2x^3 + 24x^2$

6 Simplify:

a $\frac{15k}{20k}$

b $\frac{42xy}{18y}$

c $\frac{18p^7}{9p^3}$

d $\frac{30mn^2}{9m^2n}$

7 Simplify:

a $\frac{z + 10}{10}$

b $\frac{5x^3 + 4x^5}{x}$

c $\frac{18x^3 - 12x^2 + 24}{3}$

d $\frac{20x^4 - 35x^3 + 15x^2 + 30x}{5x}$

8 Simplify:

a $\frac{6x - 16}{12}$

b $\frac{x - 4}{4 - x}$

c $\frac{3y - 45}{15 - y}$

d $\frac{m^2 - 100}{m - 10}$

e $\frac{m^2 - 1}{1 + m}$

f $\frac{a^2 - 81}{9 - a}$

g $\frac{x^2 + 4x + 4}{10(x + 2)}$

h $\frac{x^2 + 8x + 16}{(x - 5)(x + 4)}$

i $\frac{x^2 + 7x + 10}{x + 2}$

j $\frac{2x^2 + 4x - 198}{5(x - 9)}$

k $\frac{2x + 8}{x^2 + 4x}$

l $\frac{20 - 5x}{2x^2 - 8x}$

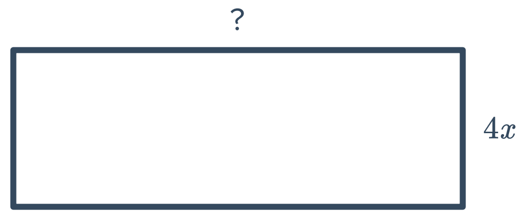
m $\frac{m^2 - 1}{2m + 2}$

n $\frac{2x^2 + 10x - 100}{2x + 20}$

o $\frac{8x^2 + 2x - 15}{(4x - 5)^2}$

Applications

9 The area of the given rectangle is $\left(4x^4 - 8x\right)$ square units. Find an expression for its length.



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